



PROTECT THE YOUNG!

One of the most important steps in freeing animals from a dependence on water was the evolution of eggs that could be laid on land. Most amphibians have to return to water to breed, but the first reptiles had eggs with lots of membranes, and later shells, to help the egg withstand dry conditions.



▲ NO NEED FOR WATER

A tortoise's eggs do not dry out due to their shells and internal membranes.



▲ **SURVIVAL** *Turtle and tortoise young spend about 6–8 weeks in their egg. All the moisture they need is contained inside the egg.*

▼ **ANCIENT** *Turtles and tortoises date back some 220 million years.*

A bit about digits

Nearly all land-dwelling vertebrates have five digits (fingers or toes) in each foot, and the same arrangement of bones in each limb. That's because they all evolved from the same ancestor—one of the early land pioneers, which happened to have five-fingered feet.

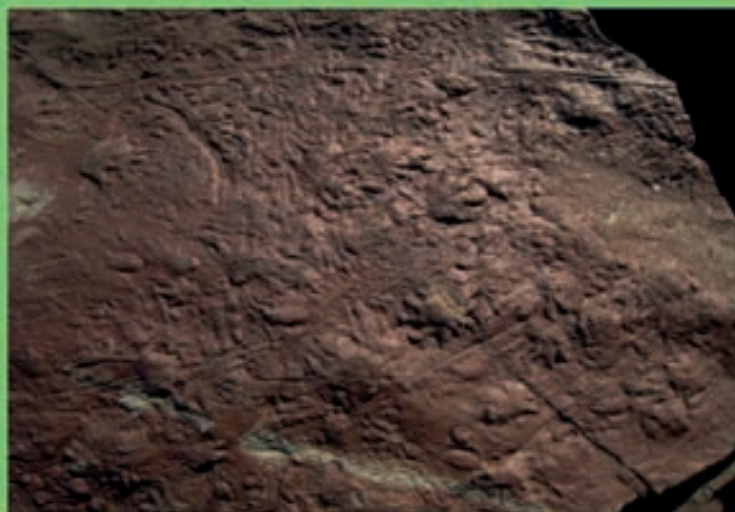


EARLY VERTEBRATES

Breathing air

Land animals take oxygen from air and have no need for the gills that fish use to extract oxygen from water. Lungs evolved in some early fish to help them gulp air at the water's surface. One group of fish retained lungs, which were crucial when they began clambering onto land. Among the first fish to clamber onto land and breathe out of water were the lungfish, some 400 million years ago.

Prehistoric lungfish were found all over the world in the Devonian Period, some 400 million years ago.



▲ **FIRST TRACKS** *These fossilized tracks were found in 2010 in Canada. They are around 318 million years old and are believed to be evidence of some of the oldest reptiles.*